
Appendix B

Equation Registry and Era-Level Calibration

B.1 Canonical Equation: The Predatory Min-Max Function

The complete formal definition of the governing optimization algorithm is reproduced here as the canonical reference. All component equations in the manuscript derive from or refer back to this canonical equation.

The Predatory Min-Max Function — Canonical Entry

Full Set-Theoretic Hierarchy:^a $E \subset P_{\text{uppet}} \subset I_{\text{buffer}} \subset I, O_{\text{racialized}} \cap I = \emptyset$.

Kernel Objective:^b

$$\max_{\{P_i\}} \mathcal{E}(t) \quad \text{subject to} \quad M_{\text{eff}}(t) = M(t) - \lambda \Phi_{\text{load}}(t) < \tau \quad (\text{B.1})$$

where the effective resistance variable is:^c

$$M_{\text{eff}}(t) = M(t) - \lambda \Phi_{\text{load}}(t) \quad (\text{B.2})$$

Suppression Envelope:^d

$$\Sigma_{\text{sup}}(t) = \psi_s(t) + \psi_m(t) + R(t) + \Phi_{\text{load}}(t) \quad (\text{B.3})$$

Crash Condition:^e

$$\frac{dM}{dt} > \frac{d\Sigma_{\text{sup}}}{dt} \iff M_{\text{eff}}(t) > \tau \quad (\text{B.4})$$

Tier Benefit Ordering:^f

$$\text{Benefit}(E) \gg \text{Benefit}(P_{\text{uppet}}) > \text{Benefit}(F_{\text{enforce}}) > \text{Benefit}(I_{\text{buffer}}) > \text{Benefit}(O_{\text{racialized}}) \quad (\text{B.5})$$

Concession Invariant: $\forall R_i \in \mathcal{R}$ (non-kinetic reforms in the 1450–2026 dataset): $\Delta \max(R_i) = 0$

Suppression Allocation: $\psi_m(t) > 0$ only when $M(t) \rightarrow \tau$; sourced from increased $O_{\text{racialized}}$ extraction.

Boundary Dynamics: The reclassification operator $\mathcal{R}(x_i)$ defined in eq. (14.14) executes whenever $K(x_i) > K_{\text{tolerated}} \vee \text{comply}(x_i) = 0$, moving x_i from I_{buffer} to O_{final} with $\psi_s(x_i) \rightarrow 0$ instantaneously; empirical instantiations include Christiana 1851, Hamburg 1876, Mulford 1967, Ruby Ridge 1992, Waco 1993, and Perdi 2026 (§14.11).

Full definition and variable legend: Section 14.7.

^aRegistry restatement of eq. (14.5) (Chapter 10).

^bRegistry restatement of eq. (1.7) (Chapter 1) and eq. (14.6) (Chapter 10). Empirical validation: see the Antebellum Kernel Optimization case study (p. 37) and the Piketty-Saez-Zucman top-decile wealth concentration analysis. Tier 3 (structural); see the Empirical Methodology chapter (p. lv).

^cRegistry restatement of eq. (14.7) (Chapter 10).

^dRegistry restatement of eq. (1.11) (Chapter 1) and eq. (14.8) (Chapter 10). Empirical validation: see the Post-1965 Backlash Wave case study (p. 52) and the ANES cross-racial solidarity proxy analysis. Tier 3 (structural); see the Empirical Methodology chapter (p. lv).

^eRegistry restatement of eq. (1.12) (Chapter 1) and eq. (14.9) (Chapter 10). Empirical validation: Bacon’s Rebellion (1676) and the Haitian Revolution (1791–1804) are the two documented cases where $M_{\text{eff}}(t) > \tau$ produced systemic kernel failure. Tier 3 (structural); see the Empirical Methodology chapter (p. lv).

^fRegistry restatement of eq. (14.10) (Chapter 10). Empirical validation: Piketty-Saez-Zucman top-decile wealth concentration, Federal Reserve SCF median household wealth by quintile, and BJS incarceration data confirming $E \cap O_{\text{final}} = \emptyset$. Tier 3 (ordinal); see the Empirical Methodology chapter (p. lv).

B.2 Electrodynamic Framework Registry

The following equations formalize the Unified Electrodynamic Framework (UEF) introduced in Chapter 0. They provide the hardware-layer translation of the set-theoretic and wave-physics formalisms deployed throughout the book. Their license to apply across substrates—the reason the same field equations govern both electrons in copper and humans in institutions—is the substrate-independence theorem established in Section 0.4.1: humans were executing these dynamics millennia before electromagnetism was formalized, and the electromagnetic vocabulary was borrowed from the prior social-power vocabulary, not the other way around. The registry below is therefore the formal apparatus the physicists left behind, applied to the system it was originally about.

Unified Electrodynamic Framework — Canonical Entry

Master Equation — The Unified Lorentz Force:

$$\vec{F}_{\text{total}} = \mathbf{Q} \vec{\mathcal{E}}_{\text{mat}} + \mathbf{Q} \left(\vec{v} \times \sum_{k=1}^N \rho_k \vec{B}_k \right) \quad (\text{B.6})$$

where $\mathbf{Q}$ is the intersectional charge multivector, $\vec{\mathcal{E}}_{\text{mat}}$ is the material-wage electric field, and $\sum \rho_k \vec{B}_k$ is the superposition of N cultural magnetic fields. The cross product $\vec{v} \times \vec{B}$ is the mathematical signature of the psychological wage: always orthogonal to velocity, so it does zero work.^a

Complex Suppression Allocation:

$$W = \psi_m + j\psi_s \quad (\text{B.7})$$

where $\psi_m \in \mathbb{R}$ is the material wage (real power) and $j\psi_s \in j\mathbb{R}$ is the psychological wage (reactive power). In polar form: $W = R e^{j\theta}$ with $R = |W| = \sqrt{\psi_m^2 + \psi_s^2}$. The phase angle θ diagnoses the operating mode: $\theta = 90$ is the default (purely psychological); $\theta > 90$ is the fascism threshold (Quadrant II).^b

Solidarity as Imaginary Cancellation:

$$W + W^* = 2\psi_m \quad (\text{B.8})$$

When the Buffer Class adopts the complex conjugate $W^* = \psi_m - j\psi_s$ and aligns with the Out-group, the imaginary status wage cancels entirely and real material power doubles.^c

Systemic Resistance (Drude Model):

$$R = \frac{m \cdot L}{n \cdot q^2 \cdot \tau \cdot A} \quad (\text{B.9})$$

The Elite engineers high resistance for the Out-group by minimizing τ (mean free time between collisions). Stop-and-frisk, welfare audits, and algorithmic flags force individuals to collide with the state before building drift velocity.^d

Enclosure Capacitance:

$$C = \frac{\epsilon A}{d} \quad (\text{B.10})$$

where ϵ is the legal dielectric (property rights, inheritance law, FHA lending), A is plate area (geographic scope), and d is distance (redlining, segregation).^e

Inductive Kickback:

$$V = -L \frac{di}{dt} \quad (\text{B.11})$$

When reform attempts to break the oppressive current ($dt \rightarrow 0$), the inductor of cultural inertia discharges stored energy as fascist backlash.^f

AC Complex Power and Real/Reactive Decomposition:

$$S = V \cdot I^* = |V| |I| e^{j\theta} = P_{\text{real}} + jQ_{\text{reactive}} \quad (\text{B.12})$$

where $P_{\text{real}} = |V| |I| \cos \theta = \psi_m$ is the material wage (real power) and $Q_{\text{reactive}} = |V| |I| \sin \theta = \psi_s$ is the psychological wage (reactive power). DC mode is the $\theta = 0$ limit; fascism threshold is $\theta > 90$.^g

N-Weighted Phase Current at the Elite Node:

$$I_E(t) = \sum_{k=1}^N \rho_k A_k \cos(\omega t + \phi_k) \quad (\text{B.13})$$

The intersectional generalization of three-phase AC: N extraction axes weighted by ρ_k (population/coupling) with phases ϕ_k tuned by the interference engine. Kirchhoff's law at the Elite node enforces extraction-current conservation under axis dropout: suppression of one A_k is rebalanced by the others weighted by $\rho_k / (\sum_{m \neq k} \rho_m)$.^h

Buffer-Class Work Theorem:

$$W_{I_{\text{buffer}}}^{\text{psych}}(t_0, t_1) = \int_{t_0}^{t_1} \mathbf{Q}(\vec{v} \times \vec{B}) \cdot \vec{v} d\tau \equiv 0 \quad (\text{B.14})$$

The systemic work performed on the Buffer Class by the psychological wage is identically zero over any interval, because $\vec{v} \times \vec{B}$ is orthogonal to $\vec{v}$ by construction. Formal proof of Du Bois's "public and psychological wage" Du Bois 1935.ⁱ

Reparations Integral:

$$\mathcal{R}(t_0, t_{\text{now}}) = \int_{t_0}^{t_{\text{now}}} \text{Re}[V_{\text{state}}(\tau) I_O^*(\tau)] d\tau \quad (\text{B.15})$$

The time integral of real power flowing from the Out-group node to the Elite node over the historical operating interval of the extraction system. Lower limits are fixed by documented system-initialization dates ($t_0 = 1444$ Portuguese, $t_0 = 1492$ Hispaniola, $t_0 = 1619$ US). Three independent peer-reviewed calibrations exist: Craemer's compounded-labor estimate yields \$14–17 trillion Craemer 2015; Darity &

Mullen’s wealth-gap-closure estimate yields \$10–12 trillion W. A. Darity and Mullen 2020; Coates’ housing/education/credit survey provides the public articulation Ta-Nehisi Coates 2014. Tier 2 (multiple peer-reviewed calibrations of the same definite integral).^j

^aRegistry restatement of eq. (0.1) (Chapter 0). Tier 3 (ordinal/structural); see the Empirical Methodology chapter (p. lv).

^bRegistry restatement of eq. (2.5) (Chapter 1). Tier 3 (ordinal/structural).

^cRegistry restatement of eq. (2.8) (Chapter 1). Tier 3 (ordinal/structural).

^dRegistry restatement of eq. (0.8) (Chapter 0). Tier 3 (ordinal/structural).

^eRegistry restatement of eq. (0.9) (Chapter 0). Tier 3 (ordinal/structural).

^fRegistry restatement of eq. (0.10) (Chapter 0). Tier 3 (ordinal/structural).

^gRegistry restatement of eqs. (0.13)–(0.14) (Chapter 0). Tier 3 (ordinal/structural).

^hRegistry restatement of eqs. (0.15)–(0.16) (Chapter 0). Tier 3 (ordinal/structural).

ⁱRegistry restatement of eq. (0.19) (Chapter 0). Tier 3 (structural/derivational).

^jRegistry restatement of eq. (0.20) (Chapter 0). Tier 2 (multiple independent peer-reviewed calibrations).

B.3 Canonical Symbol Registry

The manuscript uses a dual representation: kernel semantics (**Max/Min**) and dynamic state variables. The registry below is canonical.

Symbol	Type	Definition	Primary Mapping
$\mathcal{E}(t)$	Dynamic objective	Time-indexed extraction output routed toward E .	Dynamic analogue of max
$M(t)$	Dynamic state	Time-indexed class-coherence threat to kernel stability.	Dynamic analogue of min stress
τ	Threshold	Domestic collapse threshold for coherent class solidarity.	Failure boundary for $M(t)$ / $ S_{\text{total}} $
$\psi(t) = \psi_s(t) + \psi_m(t)$	Control variable	Suppression allocation to I_{buffer} : ψ_s (status wage, always active) + ψ_m (material wage, ≥ 0 ; deployed when $M(t) \rightarrow \tau$; sourced from increased $O_{\text{racialized}}$ extraction).	Suppression envelope component

Symbol	Type	Definition	Primary Mapping
$R(t)$	Control variable	Kinetic repression capacity via F_{enforce} .	Suppression envelope component
$\phi_{k,j}$	Phase variable	Axis-level phase shift for axis k and subgroup j .	Intersectional interference component
Φ_j	Phase aggregate	Compound subgroup phase shift: $\Phi_j = \sum_k \phi_{k,j}$.	Subgroup wave displacement
$\Phi_{\text{load}}(t)$	Dispersion index	Aggregate phase-dispersion load across active subgroups.	Suppression envelope component
$S_{\text{total}}(t)$	Wave state	Net class-solidarity signal under intersectional phase loading.	Collapse test against τ
ρ_τ	Diagnostic ratio	Proximity index $\rho_\tau = M(t)/\tau$.	Runtime-log risk indicator
$K(x_i)$	Individual state	Kinetic capacity of individual x_i : armament $\times$ demonstrated willingness to deploy against F_{enforce} .	Reclassification input; §14.11
$K_{\text{tolerated}}$	Threshold	E 's threshold for permitted Buffer-Class kinetic capacity under the current runtime; era-calibrated.	Reclassification threshold; §14.11
$\mathcal{R}(x_i)$	Boundary operator	Reclassification operator: conditional assignment of x_i to I_{buffer} or O_{final} based on $(K(x_i), K_{\text{tolerated}}, \text{comply}(x_i))$; see eq. (14.14).	Dynamic boundary function on eq. (14.13)

Symbol	Type	Definition	Primary Mapping
X_{temporal}	Mechanism set	Temporal Enclosure: the set of all sovereign debt instruments $D_{\text{sovereign}}$ whose issuance securitizes $L_{\text{future}}(O \cup I_{\text{buffer}})$ and transfers the discounted future labor surplus to E . See eq. (14.16).	Post-kinetic extraction vector; replaces spatial/kinetic enclosures as primary mechanism in mature architectures.
L_{future}	Resource variable	Aggregate future labor capacity of $O \cup I_{\text{buffer}}$: the productive output not yet performed, whose present value is securitized by $D_{\text{sovereign}}$.	Extraction target of X_{temporal} .
$D_{\text{sovereign}}$	Stock variable	Sovereign debt outstanding: the total accumulated issuance of X_{temporal} instruments; triggers eq. (18.15) when $D_{\text{sovereign}} > \bar{D}$.	Primary instrument of X_{temporal} ; accumulation clock for Interface Swap.
$\delta_E(t)$	Extraction rate	Annual extraction increment from financial repression: $\delta_E = \pi(t) - i(t) > 0$ when $r_{\text{real}} < 0$; the real purchasing power transferred per unit time from I_{buffer}/O to E . See eq. (14.17).	Stealth extraction rate; operates without kinetic enforcement.

Symbol	Type	Definition	Primary Mapping
$\mathcal{E}_{X_{\text{temporal}}}(t)$	Dynamic objective	Instantaneous temporal extraction rate: $\frac{dD_{\text{sovereign}}}{dt} \cdot \delta_E(t) \cdot \mathbf{1}[r < g]$. See eq. (14.18).	Component of $\mathcal{E}(t)$ active under $r < g$ condition; scales with sovereign debt accumulation.
$\mathcal{A}$	Architecture variable	Extraction architecture: the specific legal, monetary, and institutional substrate on which the Predatory Min-Max Function executes at a given historical moment $(\mathcal{A}_{\text{old}}, \mathcal{A}_{\text{new}})$. See eq. (18.16).	Argument of the Polymorphic Reboot Operator.
Δ_E	Continuity invariant	Change in E 's asset base across an Interface Swap: $\Delta_E = \text{Assets}(E, \mathcal{A}_{\text{new}}) - \text{Assets}(E, \mathcal{A}_{\text{old}}) \geq 0$. See eq. (18.17).	Hard constraint on every Polymorphic Interface Swap; $\Delta_E < 0$ falsifies the reboot operator.
$\mathcal{L}_{\text{Dalio}}$	Cycle index	Dalio Empire Index: 8-factor composite measuring national power (education, innovation, competitiveness, output, trade share, military, financial center, reserve currency), normalized $[0, 1]$. See eq. (14.21).	Macro-clock for the 250-year empire cycle; decline below $\mathcal{L}_{\text{Dalio}} \approx 0.75$ correlates with Demographic Paradox Limit activation.

Non-reuse constraint: E denotes the Elite class set only; $\mathcal{E}(t)$ denotes the extraction output function. ψ denotes psychological wage only. All phase operations use ϕ/Φ notation only. The symbol $\mathcal{R}$ is disambiguated by arity: $\mathcal{R}$ without arguments (as in the Concession

Invariant, §B) denotes the set of non-kinetic reforms in the 1450–2026 dataset; $\mathcal{R}(x_i)$ with an individual argument denotes the reclassification operator defined in equation (14.14). The two occupy orthogonal mathematical categories (set vs. function) and cannot be confused in well-formed expressions.

B.4 Era-Level Interference Calibration Matrix

The following ranges are bounded internal calibration intervals on a normalized [0,1] scale. They are not direct raw measurements. Their purpose is comparability across eras, not false precision.

Era / Runtime	Φ_{load}	ρ_τ	Confidence Tier	Evidence Pathway
1486 (Lisbon)	[0.10, 0.20]	[0.70, 0.85] (calibrated against Bacon/Haiti anchors; see p. lv)	Tier 2	Chapter 2 runtime + initial partition architecture + psychological wage initialization Du Bois 1935.
1676–1787	[0.15, 0.55]	(1.00, 0.75]; Anchor ($\rho_\tau = 1.0$; see p. lv)	Tier 1	Anchor ($\rho_\tau = 1.0$): Bacon crash evidence + legal codification sequence + constitutional patch dynamics.
1704–1865	[0.45, 0.60]	[0.45, 0.65] (calibrated against Bacon/Haiti anchors; see p. lv)	Tier 2	Enforcement-class consolidation + slave patrol and 13th Amendment continuity evidence.
Gendered axis	[0.35, 0.55]	[0.40, 0.60] (calibrated against Bacon/Haiti anchors; see p. lv)	Tier 2	Chapter 6 runtime + coverture + reproductive criminalization evidence.

Era / Runtime	Φ_{load}	ρ_{τ}	Confidence Tier	Evidence Pathway
1870s–1960s	[0.55, 0.72]	[0.50, 0.68] (calibrated against Bacon/Haiti anchors; see p. lv)	Tier 1	Redlining containment + two-party scaling + capture dynamics Rothstein 2017; T.-N Coates 2014.
1968–Present	[0.75, 0.92]	[0.80, 0.98] (calibrated against Bacon/Haiti anchors; see p. lv)	Tier 1	Variable Swap + carceral expansion + agenda-setting and elite-policy evidence Gilens, M. and Page, B.I 2014.
2007–2012 (Great Recession Recompile)	[0.78, 0.90]	[0.82, 0.96] (calibrated against Bacon/Haiti anchors; see p. lv)	Tier 1	Housing crash + racialized wealth loss + bottom-half balance-sheet floor + asset recovery preserving $\Delta \max = 0$ Board of Governors of the Federal Reserve System 2026; Congressional Budget Office 2024; Kochhar and Cilluffo 2017; Pfeffer, Danziger, and Schoeni 2013.

Era / Runtime	Φ_{load}	ρ_{τ}	Confidence Tier	Evidence Pathway
1992–1995 (Ruby Ridge / Waco / OKC)	[0.70, 0.85]	[0.72, 0.88] (calibrated against Bacon/Haiti anchors; see p. lv)	Tier 1	First live-action reclassification of Buffer-Class subjects under rewritten ROE; Patriot/militia misroute of kinetic response U. D. o. Justice 1994; O. o. P. R. U.S. Department of Justice 2001; Danforth 2000; Treasury 1993; Center 2015; Pitcavage 2001.
Disarmament timeline	[0.60, 0.85]	[0.70, 0.95] (calibrated against Bacon/Haiti anchors; see p. lv)	Tier 2	Kinetic-control sequence as adjunct to interference suppression across periods.
Prescriptive stress test	[0.82, 0.95]	[0.92, 1.05] (calibrated against Bacon/Haiti anchors; see p. lv)	Tier 2	Reform paradox and concession dynamics under near-threshold conditions.
Global scaling	[0.50, 0.90]	[0.55, 0.95] (calibrated against Bacon/Haiti anchors; see p. lv)	Tier 3	Imperial containment variance by region and bloc coordination UN High Commissioner for Human Rights 2026.

Era / Runtime	Φ_{load}	ρ_{τ}	Confidence Tier	Evidence Pathway
Final output	[0.85, 0.97]	[0.95, 1.08] (calibrated against Ba- con/Haiti anchors; see p. lv)	Tier 2	Book-level synthesis of terminal saturation and diminishing suppression returns.

Confidence tiers: Tier 1 = multi-source quantitative alignment in-text; Tier 2 = mixed quantitative + structural diagnostics; Tier 3 = structurally supported but data-fragmented comparative estimate.

Appendix C

Compiled Runtime Log

The following appendix concatenates all RUNTIME LOG boxes from the manuscript in chronological order, providing a single consolidated execution trace of the Predatory Min-Max Function across its five-century runtime. Each entry preserves the diagnostic format used in situ: system stress, capital output, interference state, variables deployed, and policy results.

1. RUNTIME LOG: 1486 (LISBON, PORTUGAL) — Chapter 2

System Stress (min): HIGH -- Domestic labor shortage threatening agricultural capital.
Capital (max): STAGNANT -- Sugar trade demands labor the current moral economy cannot supply.
 Φ_{load} : $[0.10, 0.20]$; $\rho_{\tau} \in [0.70, 0.85]$.
Variables Deployed: E , $O_{\text{racialized}}$, I .
Result: Extraction Algorithm initialized. *Dum Diversas* (1452), *Romanus Pontifex* (1455), *Casa dos Escravos* (1486).

2. RUNTIME LOG: 1676-1787 (VIRGINIA → PHILADELPHIA) — Chapter 3

System Stress (min): CRITICAL -- Cross-racial labor solidarity detected. Jamestown burning.
Capital (max): AT RISK -- Plantation economy destabilized by unified revolt.
 Φ_{load} : $[0.15, 0.55]$; $\rho_{\tau} > 1.00$ at crash, then $[0.60, 0.75]$ post-patch.
Variables Deployed: I_{buffer} , $F_{\text{enforce}}^{\text{proto}}$, $P_{\text{uppet}}^{\text{v1.0}}$, $W = j\psi_s$ formalized.
Result: Virginia Slave Codes (1705), Three-Fifths Compromise (1787), constitutional front-end/back-end separation.

3. RUNTIME LOG: GENDERED AXIS (FOUNDATIONAL-PRESENT) — Chapter 6

System Stress (min): MODERATE -- O_{gendered} isolated from economic and lethal autonomy.

Capital (max): HIGH -- Reproductive labor optimized through coverture and coercive control.

Φ_{load} : FRAGMENTED -- Intersection $O_{\text{racialized}} \cap O_{\text{gendered}}$ generates maximum $\alpha_{r,g}$.

Variables Deployed: O_{gendered} , $P_{\text{coverture}}$, $P_{\text{fetal_personhood}}$.

Result: Coverture (1066-1974), Buck v. Bell (1927), Dobbs (2022) Supreme Court of the United States 2022a.

4. RUNTIME LOG: 1704-1865 (AMERICAN SOUTH) — Chapter 7

System Stress (min): MODERATE -- Buffer Class pacified by $j\psi_s$. Racial partition holding.

Capital (max): EXPANDING -- Slave capitalism scaling. Human bodies as mortgageable assets.

Φ_{load} : $[0.45, 0.60]$; $\rho_\tau \in [0.45, 0.65]$.

Variables Deployed: F_{enforce} , QI (latent).

Result: Fugitive Slave Act (1850), 13th Amendment loophole (1865). max secured.

5. RUNTIME LOG: 1870s-1960s (RECONSTRUCTION → CIVIL RIGHTS) — Chapter 8

System Stress (min): HIGH -- 13th Amendment reclassified $O_{\text{racialized}}$ as citizens. Civil Rights Movement breaching the interface.

Capital (max): RESTRUCTURING -- Transitioning to indirect extraction via convict leasing, sharecropping, spatial containment.

Φ_{load} : $[0.55, 0.72]$; $\rho_\tau \in [0.50, 0.68]$.

Variables Deployed: P_{uppet} (scaled), P_{spatial} (Redlining), Capture Variable, Tweedism Filter.

Result: HOLC Redlining (1934), Civil Rights Act (1964), Voting Rights Act (1965) -- interface dismantled, kernel preserved.

6. RUNTIME LOG: v5.0 (FRACTAL INTERFERENCE ENGINE) — Chapter 12

Constraint: Binary partition alone permits periodic cross-group coherence.

Executed: Compound phase shifting across intersecting axes (race/gender/orientation/ability).

Diagnostic: $\Phi_j = \sum_{k=1}^K \phi_{k,j}$; $S_{\text{total}}(t) = \sum A_j \sin(2\pi f_{\text{class}} t + \Phi_j)$.

Result: Destructive interference maximized. Kernel stable.

7. RUNTIME LOG: 1966 (SECURITY PATCH) — Chapter 13

System Stress (min): CRITICAL -- Civil Rights breaches legal kernel. IFAS Protocol detects 91% breakthrough rate.

Capital (max): THREATENED -- Legal desegregation disrupts extraction models.

Φ_{load} : HIGH; $\rho_{\tau} \rightarrow 0.95$.

Active Patch: 1966 Moratorium, criminalizing LSD/Mescaline, recompiling Version 2.0 (Variable Swap).

Result: Epistemic Enclosure restored. Extraction migrated to race-neutral interface (War on Drugs).

8. RUNTIME LOG: 1968-1994 (THE RECOMPILE) — Chapter 13

System Stress (min): **CRITICAL** -- Demographic Paradox emerging. I_{buffer} detecting contract breach.

Capital (max): PEAK -- Carceral state fully industrialized. War on Drugs providing unlimited proxy criminalization.

Φ_{load} : $[0.75, 0.92]$; $\rho_{\tau} \in [0.80, 0.98]$.

Variables Deployed: P_{criminal} , P_{lead} , $P_{\text{deindustrial}}$, $P_{\text{BrokenWindows}}$, Universal Latent Criminality.

WARNING: min VARIABLE FAILING. System entering terminal phase.

9. RUNTIME LOG: 2007-2012 (FINANCIAL RECOMPILE) — Chapter 14

System Stress (min): HIGH -- household balance sheets collapsing; foreclosure shock destabilizing $O_{\text{racialized}}$ and lower I_{buffer} .
Capital (max): RESTORED -- state liquidity and monetary intervention stabilize financial assets and institutional balance sheets.
 Φ_{load} : $[0.78, 0.90]$; $\rho_{\tau} \in [0.82, 0.96]$.
Variables Deployed: P_{debt} , X_{temporal} , foreclosure transfer, QE asset inflation.
Result: Bottom-tier wealth destroyed; racial wealth gap amplified; Elite asset base restored; $\Delta \text{max} = 0$ preserved.

10. RUNTIME LOG: DISARMAMENT VARIABLE (FULL TIMELINE) — Chapter 16

Origin: Portuguese firearms superiority (1440s).
Feedback Loop: Arms-for-slaves trade creates exponential demand.
 Φ_{load} : $[0.60, 0.85]$; $\rho_{\tau} \in [0.70, 0.95]$.
Disarmament Protocol: Sullivan (1911) $\rightarrow$ NFA (1934) $\rightarrow$ Mulford (1967) $\rightarrow$ GCA (1968) $\rightarrow$ Hughes (1986) $\rightarrow$ Brady/AWB (1993-94) $\rightarrow$ Post-Bruen (2022-).
Pattern: Every federal gun control law correlates with $O_{\text{racialized}}$ approaching kinetic parity or P_{uppet} being kinetically corrected.

11. RUNTIME LOG: PRESCRIPTIVE STRESS TEST — Chapter 17

System Stress (min): **FAILING** -- I_{buffer} awakening. Kinetic capacity of civilian population exceeds F_{enforce} .
Capital (max): TERMINAL PHASE -- Extraction zone expanded into I_{buffer} .
 Φ_{load} : $[0.82, 0.95]$; $\rho_{\tau} \in [0.92, 1.05]$.
WARNING: A “solution” that reduces min without dismantling max is not a solution--it is maintenance.

11. RUNTIME LOG: SCALING TO IMPERIAL ARCHITECTURE — Chapter 18

System Stress (min): Historically variable. Each crash temporarily spiked min, forcing emergency patches.
Capital (max): Monotonically increasing across five centuries. Scope expanding to global field.
 Φ_{load} : $[0.50, 0.90]$; $\rho_{\tau} \in [0.55, 0.95]$ (region-dependent).
Executing: Extending 5-tier hierarchy to international system. Testing Haitian Theorem against global containment.

12. RUNTIME LOG: FINAL OUTPUT — Chapters 11–12

System Stress (min): Five centuries of runtime. $\Delta_{\max} = 0$ for every non-kinetic reform R_i .

Capital (max): O expands ($O_{\text{final}} = \text{Everyone} \setminus E$). Global containment neutralizes peripheral liberation.

$\Phi_{\text{load}}: [0.85, 0.97]$; $\rho_{\tau} \in [0.95, 1.08]$. TERMINAL SATURATION.

Variables (Complete): E/E_{global} , $P_{\text{uppet}}/P_{\text{uppet}}^{\text{global}}$, $F_{\text{enforce}}/F_{\text{enforce}}^{\text{global}}$, $I_{\text{buffer}}/I_{\text{buffer}}^{\text{global}}$, $O_{\text{racialized}}/O_{\text{global}}$.

Status: return $\vec{R}_{\text{acism}}$

Falsifiability Conditions for the Two Terminal Theorems

This appendix specifies, in precise terms, the empirical conditions under which the Concession Theorem and the Haitian Theorem would be *falsified*. The purpose is twofold. First, it demonstrates that both theorems are genuine empirical claims, not unfalsifiable definitional tautologies: there are specifiable states of the historical record that would defeat each theorem. Second, it provides a reference standard against which critics can test proposed counter-examples, and against which future evidence can be assessed as the historical record develops past 2026.

I. Falsification Conditions for the Concession Theorem

The theorem’s claim. For every non-kinetic reform R_i in the 1450–2026 dataset, $\Delta \max(R_i) = 0$: no non-kinetic reform has permanently reduced the Elite’s extraction share as measured at the system apex.

What falsification requires. A genuine falsification of the Concession Theorem must satisfy *all four* of the following conditions simultaneously:

1. **Structural reduction in extraction share:** The reform must produce a measurable, permanent decline in the top-decile’s share of total national wealth—not merely a reduction in the growth rate of inequality, but an absolute decrease in E ’s extraction share that is sustained over a minimum of three decades.
2. **No compensatory extraction elsewhere:** The reduction in the measured extraction share must not be offset by increased extraction through other mechanisms—capital flight to offshore jurisdictions, expansion of the extraction base to new populations, or substitution of financial extraction for labor extraction. The test is whether aggregate extraction by the apex (E) declined in real terms, not merely whether one extraction channel narrowed while others widened.
3. **No kinetic precondition:** The reform must have been adopted without a credible kinetic threat in the background. If the relevant reform was conceded under conditions

of rising class resistance (min approaching τ), it satisfies the Concession Theorem’s mechanism rather than refuting it—the theorem predicts that exactly such reforms will occur at exactly such moments.

4. **Sustained across threat cycles:** The reduction must persist through subsequent periods of low kinetic threat, demonstrating that the extraction kernel was structurally altered rather than temporarily suppressed. A gain that erodes once the threat subsides is evidence of the theorem, not against it.

The current state of the record. No reform in the 1450–2026 dataset satisfies all four conditions. The New Deal, the Civil Rights Act, OSHA, the Voting Rights Act, and comparable reforms in the dataset satisfy condition 1 partially and temporarily, but fail conditions 2 or 3 or both: each was adopted under elevated min conditions, and the long-run wealth concentration data shows that $\Delta \max$ returned to zero within two to three decades in every case (Figure 17.1). The theorem remains unfalsified.

What would falsify it. A concrete example: if post-2026 evidence showed that the top 10% wealth share in the United States had declined from approximately 70% (2020 baseline) to below 60% and remained there for thirty or more years following a non-kinetically-coerced reform, with no measurable compensatory extraction in other jurisdictions or asset classes, the Concession Theorem would require revision or scope restriction.

II. Falsification Conditions for the Haitian Theorem

The theorem’s claim. For all non-kinetic interventions R_i in the dataset: $\Delta \max(R_i) = 0$ (eq. 17.10). There exists at least one kinetic action K_j such that $\max(t_{\text{post}}) = 0$ locally (eq. 17.11).

What falsification requires. A genuine falsification of the strong-form Haitian Theorem must produce a documented historical case in which *all five* of the following conditions are satisfied:

1. **Intra-national racial partition:** The society in question must have an installed, operative racial partition with a functioning ψ mechanism—a buffer class holding status wages against a racially defined out-group. Systems without this architecture fall under the theorem’s scope condition (Appendix D, Section I) and cannot falsify the strong form.

-
2. **Absence of kinetic precondition:** The liberation must have been achieved without any credible kinetic threat operating in the background—no armed wing, no imminent insurrection, no defection within F_{enforce} , no external military force capable of enforcing a transition. If kinetic potential was present in any of these forms, the case resolves as the theorem’s weak form (threat without discharge) rather than a falsification.
 3. **Kernel-level liberation:** The outcome must be $\max(t_{\text{post}}) = 0$ for the prior extraction kernel, not merely an interface swap. Desegregation of public facilities while the economic commanding heights remain in Elite ownership is an interface update, not kernel termination. A falsifying case requires that the prior Elite’s extraction capacity was structurally reduced to zero—the plantation, the corporation, the financial system, whatever constitutes the extraction mechanism—not merely that its legal or political face changed.
 4. **Durable outcome:** The liberation must persist through at least one full threat cycle without requiring new kinetic pressure to maintain it, demonstrating that the kernel was dismantled rather than suppressed.
 5. **No Elite class substitution:** The liberation must not have resulted in a new extraction kernel installed by a replacement Elite—the Zimbabwe pattern (Appendix to the Boundary Conditions, Chapter 17), in which kinetic liberation produces kernel transplant rather than kernel termination, does not falsify the theorem. Falsification requires permanent architectural dismantlement, not ownership transfer.

The current state of the record. No case in the 1450–2026 dataset satisfies all five conditions. The boundary cases examined in Chapter 17—South Africa, Scandinavia, India, Zimbabwe—each fail at least one condition, and the failure in each case is structurally instructive rather than merely evidentiary. The theorem remains unfalsified in strong form.

What would falsify it. A concrete example: if a documented historical case could be established in which a society with an operative intra-national racial partition (comparable to the American 1705–2026 architecture) achieved $\max(t_{\text{post}}) = 0$ —the permanent dismantlement of the extraction kernel—through moral argument, electoral politics, litigation, or nonviolent civil disobedience, with no kinetic threat operating in any form, the Haitian Theorem’s strong form would require revision.

III. Why These Conditions Have Not Been Met

The two falsification specifications above share a common structure: they require permanent, kernel-level, non-kinetically-coerced structural change in a society with an installed racial partition. The reason this condition has not been met is, in the framework’s vocabulary, definitional: the extraction kernel is a self-reinforcing closed-loop system whose architecture—the Lyapunov stability theorem of Section 12.1.2—guarantees that every perturbation is absorbed into a descent direction of the system’s energy function. A non-kinetic intervention is, by construction, a perturbation within the system’s invariant set. Only a force that operates outside the system’s control architecture—that severs the spanning-tree topology itself rather than perturbing a node within it—can produce the structural change that falsification requires.

The theorems are not unfalsifiable because no evidence could count against them. They are unfalsified because the architectural constraint that generates their predictions is still operative. The Concession Theorem and the Haitian Theorem would be falsified by the existence of a historical counter-example meeting the specifications above. No such counter-example has been identified in the 1450–2026 dataset. The burden of proof falls on the critic to supply one.

Appendix E

Geometric Algebra and the N-Dimensional Wage

The 2D complex number $W = \psi_m + j\psi_s$ models a single imaginary axis of oppression (e.g., race). But oppression is fractal in nature and operates across multiple simultaneous, orthogonal axes. The framework must therefore be extended to higher-dimensional algebraic structures.

E.1 The Quaternion Stage: Three Intersectional Axes

For three primary imaginary axes (race, gender, sexuality), the standard complex number $a + bj$ is extended to a *quaternion*:

$$W_Q = a + bi + cj + dk \tag{E.1}$$

where i, j, k are three distinct imaginary units satisfying:

$$i^2 = j^2 = k^2 = ijk = -1 \tag{E.2}$$

Component	EE Axis	Sociological Meaning
a	Real axis	Material Wage ψ_m — wealth, land, healthcare
bi	Imaginary axis 1	Racial Status Wage (e.g., whiteness)
cj	Imaginary axis 2	Gendered Status Wage (e.g., patriarchy/maleness)
dk	Imaginary axis 3	Cisnormative/Sexuality Status Wage

Theorem E.1 (Multiplicative Intersection Compounding (Misogynoir)). *Because i, j, k are mathematically orthogonal and multiply according to quaternion rules:*

$$i \times j = k, \quad j \times i = -k \tag{E.3}$$

Intersectional oppression is not additive. For a Black woman, the intersection of the racial wage axis (i) and the gendered wage axis (j) multiplies to create a distinct third vector

k —a unique plane of systemic extraction with no simple decomposition into race alone or gender alone. This formalizes Kimberlé Crenshaw’s theory of intersectionality as a quaternion multiplication law rather than a sum.¹

E.2 The N-Dimensional Weighted Summation Equation

Beyond three axes, the framework uses *Geometric Algebra* (Clifford Algebra) over N orthogonal imaginary bases $e_1, e_2, \dots, e_N$. The full Complex Wage is:

$$W = \psi_m + \sum_{k=1}^N j_k (\rho_k \cdot \psi_{s,k}) \quad (\text{E.4})$$

where:

- ψ_m : The Real Material Wage.
- j_k : The k -th imaginary basis axis (e.g., $j_1 = \text{race}$, $j_2 = \text{gender}$, $j_3 = \text{ability}$, $j_4 = \text{sexuality}$, $j_5 = \text{neurology}$, $j_6 = \text{height/physicality}$, $\dots$).
- $\psi_{s,k}$: The baseline psychological wage on axis k .
- ρ_k : The **Demographic Weight Coefficient**—the proportion of the population for whom axis k is a primary axis of marginalization, scaled by institutional bandwidth devoted to that axis.

The Transphobia Paradox (High Voltage, Low Current)

The electrical analog of reactive power is $P = VI$ where V is voltage and I is current (population size). If ρ_k is small (small population directly experiencing a given axis), but the Elite needs large reactive power P_k from that axis to sustain the overall pacification of the Buffer Class, then:

$$P_k = V_k \cdot I_k \implies V_k = \frac{P_k}{\rho_k} \quad (\text{E.5})$$

As $\rho_k \rightarrow 0$, the required voltage $V_k \rightarrow \infty$. This mathematically predicts and explains the disproportionate legislative, media, and political bandwidth—hyper-inflated voltage—devoted to a statistically small demographic node (e.g., trans individuals).

¹This theorem is classified as Tier 3 (ordinal/structural); the quaternion formalism provides a mathematical encoding of the directional claim that intersectional oppression is qualitatively distinct from the sum of its constituent axes. See the Empirical Methodology chapter (p. lv).

The Elite pumps astronomical voltage into a small-current node to generate sufficient reactive power to distract the Buffer Class from macro-level material extraction.^a

^aThis derivation is classified as Tier 3 (ordinal/structural); it formalizes the directional claim that the Elite can achieve high reactive power from small demographic targets by intensifying the ideological voltage directed at them. See the Empirical Methodology chapter (p. lv).

E.3 Bivectors and Intersectional Space

In Geometric Algebra, multiplying two basis vectors produces a *bivector*—a distinct two-dimensional plane:

$$e_1 \wedge e_4 = e_{14} \quad (\text{E.6})$$

If e_1 is Race and e_4 is Disability, the bivector e_{14} describes the unique extraction plane of a Black disabled person. This individual does not experience $e_1 + e_4$ (additive oppression) but is located in e_{14} —a qualitatively distinct geometric zone where extraction is accelerated, enforcement violence is compounded, and systemic neglect is multiplicatively intensified.

Theorem E.2 (Fractal Consistency of Geometric Algebra). *The Geometric Algebra extension reproduces the same extraction geometry at the multi-axis level that the complex number $W = \psi_m + j\psi_s$ produces at the single-axis level. Specifically:*

- *At the single-axis level, Orthogonal Deflection is produced by the magnetic cross-product $q(\vec{v} \times \vec{B})$: the Buffer Class is deflected horizontally by the psychological wage field.*
- *At the multi-axis level, the same deflection is produced by the superposition of N cultural magnetic fields $\sum_{k=1}^N \rho_k \vec{B}_k$, each weighted by its demographic coefficient ρ_k .*

The fractal is exact: the mathematics is the same mathematics operating at different dimensional resolutions.²

²This theorem is classified as Tier 3 (ordinal/structural). See the Empirical Methodology chapter (p. lv).

Empirical Validation Index

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
Chapter 1: Redefining Racism				
eq:1.1-enclosure-collapse	Ch. 1, p. 241	T3	(ordinal)	Falsified if a population with all three outlets blocked ($e_i=1$) shows ...
eq:1.2-total-enclosure	Ch. 1, p. 257	T3	(ordinal)	Falsified if a fully blocked population ($e_1=e_2=e_3=1$) achieves coordi...
eq:1.3-conventional-causal-arrow	Ch. 1, p. 258	T3	(ordinal)	Falsified if documented evidence shows individual prejudice predating E...
eq:1.4-elite-causal-hierarchy	Ch. 1, p. 267	T3	Zurara	Falsified if racialization systems are shown to emerge without Elite ec...
eq:1.5a-kernel-optimization	Ch. 1, p. 495	T1	Piketty	Falsified if a documented period shows sustained decline in Elite extra...
eq:1.6-interface-strategy-set	Ch. 1, p. 576	T3	(ordinal)	Falsified if a documented Elite extraction system used a strategy outsi...
eq:1.7-interface-optimizer	Ch. 1, p. 608	T2	Dudziak (2000) — Cold War Civil Rights	Falsified if a historical interface transition increased net cost C_{tot} ...
eq:1.8-suppression-envelope	Ch. 1, p. 714	T2	BLS Union Membership Historical Series	Falsified if $M(t)$ consistently exceeds $\Sigma_{sup}(t)$ for extended periods wi...
eq:1.9-crash-condition-derivative	Ch. 1, p. 718	T3	(ordinal)	Falsified if a crash event (Bacon's Rebellion, Haitian Revolution) occu...
eq:1.10-effectiveness-hypothesis	Ch. 1, p. 716	T2	ANES cross-racial coalition data 1948–2020	Falsified if a crash event occurs when $M_{eff}(t) < \tau$, or if no crash occ...

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:1.11-phase-loading	Chadling 838	T3	ANES Time Series Study 1948–2020	Falsified if survey-measured cross-group solidarity remains high when $\Phi \dots$
Chapter 2: Version 1.0: Initializing the Vector (15th-Century Portugal)				
eq:2.1-racism-vector	Cart2rl. 1091	T3	(ordinal)	Falsified if documented racism operates without a state-power direction...
eq:2.2-status-suppression	Allocation	T3	Du Bois (1935) — Black Reconstruction in America	Falsified if Buffer Class loyalty is maintained without a non-material ...
eq:2.3-psi-null	China, shriv671	T3	(ordinal)	Falsified if material wages to Buffer Class are shown to be independent...
eq:2.4-roman-mother-ship	Function	T3	Bradley (1994) — Slavery and Society at Rome	Falsified if Roman enslaved population is shown to be phenotypically ho...
eq:2.5-american-cria	cial-Function	T3	Hening’s Statutes at Large (Virginia) Vol. 2	Falsified if documented American slavery assigned status based on non-p...
eq:2.6-roman-american	structural-para	T3	(ordinal)	Falsified if the structural mapping between Roman and American slavery ...
eq:2.7-three-tier	Portugal-inequality	T3	(ordinal)	Falsified if measured material outcomes show $\text{Benefit}(\text{I_buffer}) \geq \text{Benefi} \dots$
eq:2.8-church-science	Feedback-Loop	T3	Cartwright (1851) — Diseases and Peculiarities of t...	Falsified if the religious-scientific legitimization loop is disrupted fr...
Chapter 3: The Application: Bacon’s Rebellion, the Buffer Class, and the Constitutional Patch				
eq:3.1-bacon-solidarity	condition	T3	Morgan (1975) — American Slavery	Falsified if an antebellum cross-racial coalition reached $M(t) > \tau$ with...
eq:3.2-boundary-enforcement	169T3	T3	Hening’s Statutes at Large (Virginia) Vol. 3	Falsified if racial boundary enforcement persisted without legal penalt...

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:3.3-race-destruction-class	Ch. 3, p. 1539	T3	Morgan (1975) — American Slavery	Falsified if colonial-era racial coding produced class solidarity rather...
eq:3.4-three-tier-post-Bacon	Ch. 3, p. 160	T3	(ordinal)	Falsified if the post-Bacon ordering shows Buffer Class benefits exceed...
eq:3.5-kinetic-necessary-condition	Ch. 3, p. 1595	T3	Morgan (1975) — American Slavery	Falsified if documented cases show Elite extraction maintained when K_E...
Chapter 5: The Gendered Axis: Coverture, Eugenics, and the Reproductive Extraction Kernel				
eq:5.1-compliance-differential	Ch. 5, p. 2979	T2	Baptist (2014) — The Half Has Never Been Told	Falsified if $V_c(x) \leq V_r(x)$ for subjects in documented extraction regi...
Chapter 6: The Enforcement Engine: Slave Patrols, the 13th Amendment, and the Compounding				
eq:6.1-lethal-autonomy-gradient	Ch. 6, p. 2889	T2	LEOSA — 18 U.S.C. §926B statutory text	Falsified if Mapping Police Violence data shows parity in police killin...
eq:6.2-four-tier-benefit-ordering	Ch. 6, p. 2994	T3	Piketty-Saez-Zucman Distributional National Accounts	Falsified if comprehensive wealth, health, or carceral data shows Benef...
eq:6.3-thirteenth-amendment-exception	Ch. 6, p. 3080	T3	Blackmon (2008) — Slavery by Another Name	Falsified if 13th Amendment exception clause is shown to have been appl...
eq:6.4-additive-compound-policy-model	Ch. 6, p. 3144	T3	(ordinal)	Falsified if additive accumulation predicts compound policy outcomes wi...
eq:6.5-compounding-temporal-model	Ch. 6, p. 3154	T2	Hamilton & Darity (2017) — The Political Economy of...	Falsified if policy shock effects on $O^{\wedge}capacity$ are shown to be indepen...
eq:6.6-asymmetric-enforcement-multiplication	Ch. 6, p. 3172	T3	ACLU (2020) — A Tale of Two Countries: Racially Targeted...	Falsified if ACLU cannabis arrest data shows no racial disparity in pol...

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:6.7-policy-sequence-noncommutative	eq:6.7-policy-sequence-noncommutative	T3	(ordinal)	Falsified if order of policy application (P_1 then P_2 vs. P_2 then P_1...
eq:6.8-capacity-chain-1619	eq:6.8-capacity-chain-1619	T2	Darity & Mullen (2020) — From Here to Equality (rep...	Falsified if O_1619 capacity is non-significantly different from O_1450...
eq:6.9-capacity-chain-1865	eq:6.9-capacity-chain-1865	T2	Blackmon (2008) — Slavery by Another Name	Falsified if O_1865 capacity is not reduced from O_1619 as measured by ...
eq:6.10-capacity-chain-1984	eq:6.10-capacity-chain-1984	T1	Mapping Inequality — HOLC Redlining Maps	Falsified if redlining-exposed tracts show no differential wealth accum...
eq:6.11-capacity-chain-2071	eq:6.11-capacity-chain-2071	T1	ACLU (2020) — Cannabis Arrests Report	Falsified if War on Drugs enforcement shows no differential impact on O...
eq:6.12-capacity-compounding-full	eq:6.12-capacity-compounding-full	T1	ACLU (2020) — A Tale of Two Countries	Falsified if any one policy shock ($\alpha, \beta, \gamma, \delta$) is shown to have zero ma...
Chapter 7: The Containment: Pullman, Redlining, and the Wages of Whiteness (1894–1965)				
eq:7.1-pullman-door-latch	eq:7.1-pullman-door-latch	T3	Roediger (1991) — The Wages of Whiteness	Falsified if a documented case shows I_buffer achieving collective gain...
Chapter 8: Tweedism and the Puppet Class: The Algorithmic Filter on Democracy				
eq:8.1-vertex-partition-graph	eq:8.1-vertex-partition-graph	T3	(ordinal)	Falsified if leader/follower partition fails to predict convergence in ...
eq:8.2-tier-to-graph-mapping	eq:8.2-tier-to-graph-mapping	T3	(ordinal)	Falsified if the tier-to-graph mapping produces inconsistent prediction...
eq:8.3-fractional-agent-dynamics	eq:8.3-fractional-agent-dynamics	T3	Li et al. — Mittag-Leffler stability results (mathe...	Falsified if integer-order dynamics ($\alpha=1$) predict social-mobility conve...
eq:8.4-compounding-chain-formal	eq:8.4-compounding-chain-formal	T2	Federal Reserve SCF racial wealth data	Falsified if multiplicative chain prediction diverges significantly fro...

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:8.5-stationary-leader-condition	Gly. 8, sheet 16	T3	(ordinal)	Falsified if documented Elite actors show systematic response to u_i (u...
eq:8.6-proximity-condition	Gly. 8, sheet 18	T3	(ordinal)	Falsified if social-affinity graph connectivity is shown to be independ...
eq:8.7-navigation-components	Gly. 8, sheet 20	T3	(ordinal)	Falsified if goal and constraint functions do not bound follower trajec...
eq:8.8-navigation-function	Gly. 8, sheet 22	T3	(ordinal)	Falsified if ϕ_i fails to produce convergence in any graph satisfying t...
eq:8.9-gradient-control-law	Gly. 8, sheet 28	T3	(ordinal)	Falsified if negative gradient of ϕ_i fails to drive convergence in any...
eq:8.10-convex-hull-convergence	Gly. 8, sheet 36	T3	(ordinal)	Falsified if followers fail to converge to Elite convex hull in any gra...
eq:8.11-mittag-leffler-stability	Gly. 8, sheet 40	T3	Chetty et al. (2018) — Race and Economic Opportunit...	Falsified if social-mobility data for $\alpha \in (0,1)$ shows exponential rathe...
eq:8.12-class-alignment-base-wave	Gly. 8, sheet 46	T4	Google Trends	Falsified if class alignment signal is not measurable as a distinct fre...
eq:8.13-subgroup-compound-phase	Gly. 8, sheet 48	T2	Google Trends	Falsified if compound phase Φ_j fails to predict subgroup political ali...
eq:8.14-net-solidarity-signal	Gly. 8, sheet 50	T1	Google Trends	Falsified if net solidarity signal cannot be measured as distinct from ...
eq:8.15-solidarity-class-condition	Gly. 8, sheet 52	T1	Google Trends	Falsified if $M(t)$ remains above τ even when S_{class} approaches zero acr...
eq:8.16-interference-control-objective	Gly. 8, sheet 54	T1	Google Trends	Falsified if observed interference engine output increases S_{class} rath...

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:8.17-circular-dispersion-op	Ch. 8, 1968	F2	ANES Time Series — cross-group solidarity items	Falsified if circular dispersion of phase values fails to predict cross. . .
eq:8.18-tweedish-agenda-path	Ch. 8, 1844	T1	Gilens & Page (2014) — Testing Theories of American. . .	Falsified if Gilens-Page analysis shows median voter preferences predic. . .
Chapter 9: The Recompile: COINTELPRO, the Variable Swap, and the War on Drugs (1968–1971)				
eq:9.1-lead-crime-compounding	Ch. 9, 6344	T1	Reyes (2007) — Environmental Policy as Social Policy	Falsified if Reyes (2007) blood-lead time-series fails to predict crime. . .
eq:9.2-epistemic-suppression-val	Ch. 9, 6344	F2	Kitman (2000) — The Secret History of Lead (The Nat. . .	Falsified if documented lead industry cover-up timeline shows no correl. . .
eq:9.3-multi-vector-lead-exposure	Ch. 9, 6344	F1	EPA Air Quality Monitoring Data	Falsified if cumulative lead exposure from all three vectors is not sig. . .
eq:9.4-redlining-containment-field	Ch. 9, 6344	F1d	Mapping Inequality HOLC maps	Falsified if HOLC map grades show no statistically significant correlat. . .
eq:9.5-school-funding-property-value	Ch. 9, 6337	F1	NCES Public School Finance Survey	Falsified if school funding is shown to be independent of property valu. . .
eq:9.6-infrastructure-quality-funding	Ch. 9, 6337	F2	NCES School Facility Condition Survey	Falsified if school infrastructure quality is shown to be unrelated to . . .
eq:9.7-school-lead-exposure-inverse	Ch. 9, 6337	F1	EPA 3Ts for Reducing Lead in Drinking Water in Scho. . .	Falsified if school infrastructure quality is shown to be unrelated to . . .
eq:9.8-community-capacity-lead-reduction	Ch. 9, 6337	F2	Aimer & Currie (2019) — Lead and Juvenile Delinquen. . .	Falsified if community economic capacity is shown to be independent of . . .
eq:9.9-property-value-capacity-feedback	Ch. 9, 6337	F1	Mapping Inequality HOLC maps + ACS property value d. . .	Falsified if property values in subsequent periods are shown to be inde. . .

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:9.10-highway-lead-spatial-coordination	Chadwick	T1	Mapping Inequality DSL (Nelson et al.) — HOLC grade...	Falsified if HOLC-D tracts show no excess firearm or lead burden relati...
Chapter 10: The Full Algorithm: Demographic Paradox, Cannibalization, and the 5-Tier Reveal				
eq:10.1-outgroup-racialized-subset	Chadwick	T8	US Census Bureau demographic data	Falsified if O_racialized is shown to be coextensive with O rather than...
eq:10.2-buffer-class-size-linkage	Chadwick	T2	Pew Research Center — America's Shrinking Middle Cl...	Falsified if Buffer Class size is shown to be expanding rather than con...
eq:10.3-cannibalization-equation	Chadwick	T3	Saez-Zucman wealth share series	Falsified if documented Elite extraction rate increases without corresp...
eq:10.4-predatory-inheritance-definition	Chadwick	T1		Falsified if the five-tier hierarchy fails to map onto any contemporary...
eq:10.5-kernel-objective-canonization	Chadwick	T1	Piketty-Saez-Zucman Distributional National Accounts	Falsified if a documented period shows sustained Elite wealth-share dec...
eq:10.6-effective-resistance-variable	Chadwick	T1		Falsified if M_eff fails to capture the difference between raw class co...
eq:10.7-suppression-envelope-canonization	Chadwick	T2		Falsified if documented suppression tools outside this set constitute t...
eq:10.8-crash-condition-canonization	Chadwick	T3		Falsified if crash events occur in documented cases without $M_eff(t) > \dots$
eq:10.9-tier-benefit-bracketing-canonization	Chadwick	T3		Falsified if any two-tier comparison fails to show stated ordering in c...
eq:10.10-system-objective-extraction-ratio	Chadwick	T3		Falsified if documented extraction systems show this ratio declining wi...

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:10.11-system-stability-algorithm	Catala 17405	T3	(ordinal)	Falsified if system stability is maintained when extraction/resistance ...
eq:10.12-o-final-construction	Ch. 10, 16828	T1	Bureau of Justice Statistics incarceration by race ...	Falsified if modern mass-incarceration demographics fail to reproduce t...
eq:10.13-reclassification-operator	Ch. 10, 7734	T3	(ordinal)	Falsified if the operator fails to predict observed ideological sorting...
eq:10.14-kinetic-power-distribution	Ch. 10, 7768	T2	Pew Research Center — Gun Ownership Survey 2017	Falsified if kinetic power is not concentrated in F_enforce as a propor...
eq:10.15-temporal-enclousure-definition	Ch. 10, 7800	T3	Reinhart-Sbrancia 2015 (Economic Policy)	Falsified if sovereign debt accumulation is demonstrably unrelated to p...
eq:10.16-financial-repression-condition	Ch. 10, 7825	T1	FRED 10Y Treasury Constant Maturity Rate (GS10)	Falsified if: (1) real interest rates remain positive throughout the 19...
eq:10.17-temporal-extraction-rate	Ch. 10, 7825	T4	FRED: Federal Debt Total Public Debt as % of GDP (G...	Falsified if the indicator function $1[r < g]$ is inactive during documen...
eq:10.18-psi-degradation-function	Ch. 10, 7905	T1	FRED: Real Median Hourly Compensation (B4701C0A052N...	Falsified if: (1) Chow test at 1971 shows no statistically significant ...
eq:10.19-saeculum-phase-map	Ch. 10, 7909	T2	Strauss & Howe, The Fourth Turning (1997)	Falsified if: (1) the generational archetype sequence does not follow t...
eq:10.20-dalio-extraction-bound	Ch. 10, 7950	T2	Maddison Project Database 2020 (GDP, trade output)	Falsified if: (1) the reconstructed US Empire Index does not show a pea...
Chapter 11: The Kinetic Guarantee: Arms Asymmetry, the Second Amendment, and the Disarmament				
eq:11.1-extraction-precondition	Ch. 11, 7992	T1	Primary legislation texts (sullivanlaw, nfa1934, mu...	Falsified if a documented extraction system maintained stability with L...

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:11.2-disarmament-legislation-path	Gentile, 1992	T1	Primary legislation and ruling texts (11 events)	Falsified if documented gun-control legislation passed under high Φ_{loa} ...
eq:11.3-restriction-precedent-cumulation	Chion, 1992	T1	Primary legislation and ruling texts (11 events)	Falsified if any documented legislative or ruling event reduced cumulat...
eq:11.4-restriction-scope-expansion	Chion, 1992	T1	Primary legislation texts and enforcement records	Falsified if the historical record showed restriction scope declining o...
Chapter 12: The Contradiction: Why Reform Serves the Algorithm				
eq:12.1-reform-absorption-mechanism	Chion, 1985	T1	Piketty-Saez top income share data	Falsified if a documented reform reduces Elite extraction share (max) r...
eq:12.2-concession-theorem	Chion, 1990	T2	Piven & Cloward (1977) — Poor People's Movements	Falsified if Elite-initiated reforms consistently exceed the minimum re...
eq:12.3-judiciary-discrimination-detection	Chion, 1997	T3	Washington v. Davis	Falsified if judicial detection function produces false positives for p...
eq:12.4-proxy-discrimination-equivalence	Chion, 1997	T3	Alexander (2010) — The New Jim Crow	Falsified if P_proxy achieves racial separation at rates significantly ...
eq:12.4a-judicial-double-agent	Chion, 1997	T2	Local SCOTUS markdown corpus	Falsified if the SCOTUS corpus does not separate civil-rights/voting ca...
eq:12.5-haitian-theorem-nonkinetic	Chion, 1985	T1	James (1938) The Black Jacobins	Falsified if any documented non-kinetic reform achieves $\Delta_{\text{max}} < 0$ (susta...
eq:12.6-haitian-theorem-kinetic	Chion, 1985	T1	James (1938) The Black Jacobins	Falsified if any kinetic event fails to produce at least temporary loca...
eq:12.7-gendered-out-group-set	Chion, 1997	T3	(ordinal)	Falsified if O_gendered is shown to be structured differently from O_ra...

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:12.8-intersectional-domination-partition	Crit12-10012	F1	Crenshaw (1989) — Demarginalizing the Intersection ...	Falsified if intersection $O_racialized \cap O_gendered$ does not show compo...
eq:12.9-multiplicative-intersection-compounding	Crit12-9607	F2	AJWindinggap data by race and gender	Falsified if Black women's wealth gap from white men is not multiplicat...
eq:12.10-nonviolence-mandate-function	Crit12-9709	F8	Carson (1981) — In Struggle: SNCC and the Black Awa...	Falsified if documented liberation movements adopting nonviolence achie...
Chapter 13: The Global Containment Field: Scaling the Algorithm				
eq:13.1-global-compounding-capacity	Crit13-9905	F2	World Bank development indicators — GDP per capita ...	Falsified if post-colonial nation's capacity trajectory shows exponenti...
eq:13.2-global-predatory-minimax	Crit13-9973	T3	Klein (2007) — The Shock Doctrine	Falsified if documented international economic relationships fail to ma...
eq:13.3-externalized-enforcement-envelopes	Crit13-9903	F3	UN MINUSTAH mission records	Falsified if Haiti's domestic coercive capacity is shown to be structur...
eq:13.4-killick-extraction-equation	Crit13-10024	F10	NYT 'The Ransom' (2022)	Falsified if the extraction equation fails to predict subsequent debt i...
eq:13.5-legitimation-constraint-set	Crit13-10039	F8	(ordinal)	Falsified if E_global actors consistently operate outside their stated ...
eq:13.6-authoritarian-precondition	Crit13-10050	T8	(ordinal)	Falsified if E_global achieves extraction without satisfying authorizat...
eq:13.7-coercion-legitimacy-frontier	Crit13-10056	T8	(ordinal)	Falsified if observed coercion levels in 'voluntary' agreements consist...
eq:13.8-firm-extraction-threshold	Crit13-10062	F3	(ordinal)	Falsified if E_global achieves extraction with deployed force below F^* ...

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:13.9-infeasible-consistent-set	Chile, 1977	T3	(ordinal)	Falsified if E_global achieves both extraction success and legitimation...
eq:13.10-asymmetric-leverage	Chile, 19100	T2	Firmin (1885) — De l'égalité des races humaines	Falsified if a weaker actor's invocation of E_global's legitimation arc...
eq:13.11-legitimation-abandonment	Chile, 1907	T3	(ordinal)	Falsified if E_global maintains legitimation constraints even when extr...
eq:13.12-global-containment-strategy	China, 19423	T2	(ordinal)	Falsified if global containment strategies differ structurally from the...
eq:13.13-un-voting-distribution	Chile, 19162	T3	UN Digital Library — UNSC vote records	Falsified if UN Security Council voting distribution fails to map onto ...
eq:13.14-imperial-core-collapse	Chile, 19208	T1	World Bank Development Indicators — GDP trajectory ...	Falsified if documented rising powers (China, OPEC, Asian Tigers) achie...
eq:13.15-interface-swap-trigger	Chile, 19280	T1	Federal Reserve H.4.1: Historical Gold Stock Data (...)	Falsified if the gold-cover ratio was still above 80% at the time of th...
eq:13.16-polymorphic-reboot-operator	Chile, 19310	T2	Gowa 1983 (Closing the Gold Window)	Falsified if Assets(E, A_new) < Assets(E, A_old) across any of the thre...
eq:13.17-elite-asset-concavity	Chile, 19320	T1	WID.world: Top-0.1% US Wealth Share 1913-present (P...	Falsified if the top-0.1% US wealth share declines STRUCTURALLY (> 3 pe...
Chapter 14: The Algorithmic Epoch: Real-Time Subjugation and the Necessity of the Counter-V				
eq:14.1-algo-prior-inherence	Chile, 19757	T2	Angwin et al. (2016) — Machine Bias (ProPublica)	Falsified if ML systems trained on pre-2024 US data show no racial disp...
eq:14.2-realtime-proximity-estimate	Chile, 19777	T2	Twitter/X political affinity graph data (academic d...	Falsified if real-time social-affinity graph G(t) shows no correlation ...

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:14.3-orthogonal-vector-injection	Challenger 10793	T3	(ordinal)	Falsified if documented interference engine operations increase class-a...
eq:14.4-correction-signal-damping	Challenger 1807	T3	(ordinal)	Falsified if scale-accurate counter-signals are shown to reduce identit...
eq:14.5-optical-center-loss	Challenger 1820	T3	Pew Research — Police-community relations by race	Falsified if documented perception studies show O's population maintain...
eq:14.6-decoy-transfer-coefficient	Challenger 1083	T2	(pending)	Falsified if documented cases show kinetic class energy passing to E at...
eq:14.7-ddos-bandwidth-censoring	Challenger 1055	T2	Armed Conflict Location & Event Data (ACLED) — US p...	Falsified if F_enforce simultaneously suppresses $n > \Gamma$ nodes in any doc...
eq:14.8-armed-swarm-bandwidth	Challenger 1080	T2	(pending)	Falsified if F_enforce simultaneously neutralizes armed nodes with forc...
eq:14.9-kinetic-crash-site-asymmetry	Challenger 1087	T2	Pew Research Center — Gun Ownership Survey 2017	Falsified if gun-ownership data shows no racial asymmetry in kinetic ca...
eq:14.10-sustained-siege-cost	Challenger 1000	T2	(pending)	Falsified if F_enforce besieges multiple simultaneous armed nodes with ...
eq:14.11-escalation-ceiling-tiers	Challenger 1010	T3	(ordinal)	Falsified if F_enforce escalation tiers do not follow the predicted ban...
eq:14.12-inclusion-prediction	Challenger 1046	T2	Bureau of Labor Statistics contingent worker survey	Falsified if data shows $V(x,t) \leq R(x,t)$ for majority members of I_buffer...
eq:14.13-terminal-interface-swap	Challenger 1050	T3	(ordinal)	Falsified if documented late-stage extraction shows I_buffer benefits m...

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:14.14-liability-phase-optimization	City-Phase-Optimization	T3	(ordinal)	Falsified if documented terminal-phase extraction shows E maintaining I...
eq:14.15-noise-spectrum-index	Spectrum-Index	T2	ACLED — US social movements 2011–2020	Falsified if noise-spectrum index of a documented swarm fails to predic...
eq:14.16-agnostic-swarm-load	Cic-swarm-load	T3	(ordinal)	Falsified if documented multi-node mobilizations with ideological diver...
eq:14.17-botnet-load-theorem	Load-theorem	T2	ACLED — 2020 US protest dataset	Falsified if enforcement grid contains a swarm with $n > \Gamma_{\text{armed nodes a...}}$
eq:14.18-ideological-routing-operator	Idcl-routing-operator	T3	(ordinal)	Falsified if nodes in documented distributed mobilizations fail to rout...
eq:14.19-zugzwang-payoff-function	Zugzwang-payoff-function	T2	(pending)	Falsified if Elite actors systematically choose attack over stand-down ...
eq:14.20-defection-taskade	Cionl-taskade	T2	James (1938) — The Black Jacobins (Polish Legion de...	Falsified if documented enforcement nodes fail to defect when community...
eq:14.21-empathy-permeability	Cy-permeability	T2	James (1938) — The Black Jacobins	Falsified if trauma overlap between enforcement and out-group nodes fai...
eq:14.22-semantic-overwrite	Cic-overwrite	T3	Haitian Constitution of 1805 (primary source)	Falsified if 1805 Haitian Constitution's ideological redefinition faile...

Chapter 17: Conclusion

eq:17.1-racism-as-institutional-vector	As-institutional-vector	T3	(ordinal)	Falsified if documented cases of marginalized-group prejudice achieve s...
eq:17.2-o-terminal-expansion	Omial-expansion	T2	US Census Bureau 2020 demographic projections	Falsified if documented demographic trends show O contracting rather th...

Chapter 19: Equation Registry and Era-Level Calibration

Equation label	Case study location	Tier	Primary data source	Falsification (truncated)
eq:19.1-kernel-Objective-Registry-entry (ordinal)	Condition 11717	F8		Redundant with eq:5 and eq:56; falsified by the same conditions as thos...
eq:19.2-suppression-envelope-Registry-entry (ordinal)	Condition 11701	F8		Redundant with eq:8 and eq:58; falsified by the same conditions as thos...
eq:19.3-crash-Condition-Registry-entry (ordinal)	Condition 11705	F8		Redundant with eq:9 and eq:59; falsified by the same conditions as thos...
eq:19.4-tier-ordering-Registry-Entry (ordinal)	Condition 11729	F8		Redundant with eq:60; falsified if any tier ordering is reversed in com...

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